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Anticipation and tracking of pulsed resources drive population dynamics in eastern chipmunks

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Abstract. Pulsed systems are characterized by boom and bust cycles of resource production that are expected to cascade through multiple trophic levels. Many of the consumers within pulsed resource systems have specific adaptations to cope with these cycles that may serve to either amplify or dampen their community-wide consequences. We monitored a seed predator, the eastern chipmunk (*Tamias striatus*), in an American beech (*Fagus grandifolia*) dominated forest, and used capture–mark–recapture analyses to estimate chipmunk vital rates and relate them to interannual variation in beech seed production. The summer activity and reproduction of adults anticipated autumn beech production, with high activity and intense reproduction occurring in summers prior to beech masts. Chipmunks also reproduced every spring following a beech mast. However, adult survival was independent of beech production. In contrast, juvenile survival was lower in years of mast failure than in years of mast production, but their activity was consistently high and independent of beech production. Population growth was strongly affected by the number of juveniles and therefore by beech seed production, which explains nearly 70% of variation in population growth. Our results suggest that a combination of resource-dependent reproduction and variable activity levels associated with anticipation and response to resource pulses allows consumers to buffer potential deleterious effects of low food abundance on their survival.

Key words: American beech; anticipatory reproduction; capture–mark–recapture; eastern chipmunk; Fagus grandifolia; *life-history trade-offs; masting; reproduction skipping; resource tracking; resource–consumer interactions;* Tamias striatus.

Introduction

The dynamics of many ecosystems are characterized by pulsed resources, leading to cascading demographic consequences on consumer populations at multiple trophic levels (reviewed in Ostfeld and Keesing 2000, Yang et al. 2010). Resource pulses are typically defined as infrequent episodes of resource availability that are large in magnitude and short in duration (Yang et al. 2008). Insect outbreaks and the synchronous, but intermittent, production of large seed crops by plant populations (i.e., mast) are well-known examples of resource pulses (reviewed in Yang et al. 2010). Population dynamics of consumers are particularly sensitive to variation in food abundance (Jones et al. 1998, Yang et al. 2010). For example, the reproductive events of many species have frequently been shown to track pulsed resources (Ostfeld et al. 1996, Clotfelter et al. 2007, Yang et al. 2010). Furthermore, the prolonged absence of resources between pulse events can constrain key fitness components (e.g., survival and reproductive rates) and contribute to population collapse (reviewed in White 2008). A resulting key feature of the consumer–resource interaction is a strong lagged demographic response of consumer communities to food abundance (Jones et al. 1998, Ostfeld and Keesing 2000, Yang et al. 2010). Such a lag may cause juveniles to enter the population when food abundance is low, potentially reducing their survival compared to juveniles born prior to or during resource pulse events (White 2008). For short-lived consumers, the main determinant of population persistence over time is usually early adult fertility and juvenile recruitment rather than prolonged adult survival (Heppell et al. 2000; but see McAdam et al. 2007). These consumers should thus aim to adopt life-history strategies that circumvent lagged demographic effects (Boutin et al. 2006, Marcello et al. 2008).

A common approach to studying the consequences of pulsed resources on consumer demography is to examine how vital rates are affected by past or current food availability. Although resource pulses have repeatedly been shown to affect consumer reproduction, other vital rates such as survival are essential to document, because they reflect the capacity for adults to persist until the next resource pulse (Ruf et al. 2006). To favor survival during

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prolonged periods of low food abundance, individuals could adopt a conservative life-history strategy by delaying reproduction and allocating more energy to maintenance (Stearns 1992, Ruf et al. 2006). However, reproducing only in response to resource pulses may be problematic for short-lived species, especially in the context of rare or irregular resource pulse events.

In addition to the classical resource-tracking strategy, cues allowing resource pulse anticipation by consumers could also be important drivers of their life-history strategies (Boutin et al. 2006, Marcello et al. 2008). For instance, rodents such as white-footed mice (*Peromyscus leucopus*) are able to anticipate spring cicada (*Magicicada* spp.) outbreaks (Marcello et al. 2008) and red squirrels (*Tamiasciurus hudsonicus* and *Sciurus vulgaris*) are able to anticipate tree masting events (Boutin et al. 2006). Anticipatory life histories allow these species to synchronize juvenile emergence with peak food abundance, which contributes to juvenile survival (White 2008). Squirrels, like many other rodent species, also reproduce following masting events (Boutin et al. 2006); thus, mast anticipation provides an additional reproductive opportunity for a single masting event (Boutin et al. 2006). Given that anticipation of resource pulses seems possible in rodents, it should also influence other aspects of the ecology of these species, such as seasonal activity patterns. For example, during periods of low resource availability, the costs of foraging (e.g., energy, injury, and predation) may outweigh the potential energetic gain. In such conditions, animals might reduce time spent foraging for food and rely on previously stored resources while they remain in the safety of their refuges. However, few studies have corroborated the findings of Boutin et al. (2006) on anticipatory reproductive behavior (e.g., Falls et al. 2007, Marcello et al. 2008), and no study has examined whether the capacity to reproductively anticipate resource pulses does or does not extend to the regulation of activity patterns and its resulting effect on adult survival.

Eastern chipmunks (*Tamias striatus*) are small, forest-dwelling mammals that consume and hoard seeds produced by masting deciduous trees (Snyder 1982). Chipmunks display variable levels of aboveground activity (Munro et al. 2008) and can reproduce during two distinct reproductive seasons: early spring, summer, or both (Bergeron et al. 2011). Although chipmunks consume a variety of food items, their larder is mostly composed of seeds from dominant masting deciduous trees (Elliott 1978, Snyder 1982, Humphries et al. 2002). We thus expect that mast anticipation, in addition to typical reproductive responses to seed availability, is a major driver of chipmunk activity and life-history strategies. We hypothesize that mast seeding of American beech (*Fagus grandifolia*), the dominant canopy tree and the main source of storable food items for chipmunks at our study site, is the main determinant of both reproduction and aboveground activity patterns of chipmunks. We predict that spring reproduction occurs only in response to a beech mast during the preceding autumn. We also expect that if chipmunks are able to anticipate pulsed resources, they should reproduce in the summer of mast years only (for juvenile emergence in autumn). Also, we predict that autumn beech mast anticipation regulates summer aboveground activity, such that chipmunks will remain underground during summers preceding beech mast failure. Finally, we predict that if chipmunks adopt conservative life-history strategies, adult survival should be partly buffered against environmental variation due to the ability of females to skip reproductive events (Ruf et al. 2006). We therefore expect low temporal variability in adult survival that is independent of beech seed production.

Methods

Study site and data collection.—We monitored an eastern chipmunk population from 2005 through 2010 on a 25-ha study site located in a deciduous forest of southern Québec, Canada (45°05′ N, 72°26′ W). The entire study area was marked with numbered stakes at 20-m intervals. We designated a fixed trapping grid inside a 250 m radius circle and used every second grid stake as a permanent trap location, for a total of 228 locations. We trapped chipmunks, using Longworth traps checked at 2-h intervals, daily from 08:00 hours until dusk during their active period, every week from the end of April to October. At each capture, we weighed all individuals to the nearest 1 g using a Pesola balance and determined their reproductive status. Females were considered reproductively active if captured with a swollen vulva or developed mammae. Every new animal was aged, sexed, and uniquely marked with ear tags and a Trovan PIT tag.

Juveniles were differentiated from adults based on either their mass at emergence (juveniles < 80 g) or the absence of darkened scrotum or developed mammae for individuals > 80 g (Careau et al. 2010). Chipmunks can reach sexual maturity at six months of age (Snyder 1982). At their first capture, juveniles caught before 1 August were assigned to spring reproduction, whereas juveniles caught after 1 August were assigned to summer reproduction. Juveniles born from spring reproduction were classified as adults after 1 August of the same year, whereas juveniles born in summer were classified as adults in the following spring. Juveniles were considered to have recruited if they survived beyond the time point when they were classified as adults.

We quantified autumn seed production of beech using 30 plastic buckets (with 0.06-m^2 opening) placed under the canopy of 30 beech trees having a circumference at breast height of $\geq$10 cm and evenly distributed on the trapping grid (Landry-Cuerrier et al. 2008, Munro et al. 2008). To avoid seed removal by seed consumers, buckets were more than 40 cm deep, installed 1 m away from the trunk of the target tree, and mounted on two smooth metal posts 50 cm above ground level. We collected bucket contents twice between September and November. All seed coats were opened and the nuts (the only part consumed by the chipmunks) were weighed. We considered the average mass of the nuts collected (in grams per square meter) as a proxy of beech seed production (see also Landry-Cuerrier et al. 2008). The amount of beech seeds collected in the buckets corroborated our qualitative observations of mast/non-mast events, as masts are conspicuous when they occur, leaving the forest floor littered with seeds (Munro et al. 2008).

Statistics.—We calculated average reproductive success as the ratio of the number of juveniles and the number of adult females captured prior to, or after 1 August, for spring and summer reproduction, respectively. We used a nonparametric Spearman correlation test to assess the relationship between average summer reproductive success and beech seed production during the corresponding autumn. Because the amount of seeds remaining in burrows at emergence in spring could not be quantified, we used the quantity of seeds produced in the preceding autumn as an index of seed availability during spring reproduction.

We estimated survival and recapture rates of adult and juvenile chipmunks using the maximum likelihood procedure from capture–recapture histories with the program MARK 6.0 (White and Burnham 1999); see Appendix A. To model capture histories, we divided the annual active period of chipmunks into two discrete periods of three months each, pertaining to the seasons associated with potential juvenile emergence (Fig. 1). The first season (early active, E) spanned from spring emergence (late April/May) to the end of July; the second season (late active, L) started on 1 August and lasted until all chipmunks retreated in their burrows for the winter (as late as the end of October). Survival estimates are calculated based on the time elapsed between the median dates of each season (E, 15 June; L, 15 September). Hence, the capture histories allowed us to estimate local survival rates (ϕ) during the active period (from E to L, 3 months) and inactive period (from L to E, 9 months), and to estimate recapture rates (p) at E and L. We used Cormack-Jolly-Seber models to estimate ϕ and p probabilities for a total of 11 seasons (occasions) between spring 2005 and spring 2010 (Lebreton et al. 1992). All individuals captured at least once during a given season were included in the analyses. Model selection in MARK was based on estimates transformed to six-month periods, representing chipmunk periods of activity (May to October) and inactivity (November to April); see Appendix A.

We also estimated population size at time $t+1$ as the product of population size at t and survival rates, and 95% confidence intervals as the mean values obtained from 2500 Monte Carlo iterations, in which different survival rates were randomly selected (at each iteration) within the confidence interval boundaries provided by MARK (Appendix B). We calculated population growth as the ratio between population size at time $t+1$ and population size at t (Coulson et al. 2008). We used linear models to assess both the relationship between population growth and beech seed production and the relationship between population growth and the number of resident adults, immigrants, recruiting juveniles, and juveniles. We used R 2.12.2 (R Development Core Team 2011) to perform analyses not accounted for by MARK. If not otherwise stated, all means are presented $\pm$ SE.

Results

Beech masting and its effects on reproduction.—We observed extreme annual fluctuations of beech seed production over our study area (complete mast failures in 2005 and 2007; masts in 2006 and 2008; see Fig. 1). Chipmunks did not reproduce (average reproductive success = 0) prior to and after complete mast failures (Fig. 1). However, during mast years, chipmunks mated in June before beech seeds were available for consumption, and in March following masting events (Fig. 1). Reproduction occurred twice during the 2009 active season when seed production was small ($0.15 g/m^2$), but there was no reproduction in the following spring 2010 (Fig. 1). Over all 11 seasons, there was a positive correlation between beech seed production and the number of juveniles per adult female (Spearman's rho = 0.81, $P = 0.002$).

Beech masting effects on aboveground activity and survival.—An index of adult aboveground activity, measured by recapture probabilities, also varied greatly among seasons and sexes, although there was no consistent (i.e., additive) difference between sexes through time (Fig. 2a; Appendix C: Table C1a). Adult aboveground activity was usually high during early active seasons, independent of mast, and also during late active seasons when there was mast (Fig. 2a). However, male and female activity was much lower in late active seasons coinciding with mast failures, compared to the other occasions (Fig. 2a; Appendix C: Fig. C1). The strongest decline was observed in autumn 2007, when only 5/143 adult chipmunks known alive were active aboveground after July (Fig. 2a; Appendix C: Fig. C1). Juveniles were always highly active in the season following their birth when they recruited in the adult population (Fig. 2b), independent of beech seed production, even during the late active period of 2007 when virtually all adults were inactive (Fig. 2a, b).

The most parsimonious model of local adult survival ($n = 493$ capture histories) included only time-dependent effects on survival (Fig. 2c; Appendix C: Table C1a). Survival rates varied from 0.48 to 0.90 (Fig. 2c), independently of beech seed production and seasons. Models with beech seed production, models with seasonal effects, and models accounting for interactions between these two environmental components did not provide support to the data compared to the fully time-dependent model (Appendix C: Table C1a). Also, there was no strong support for a sex effect on survival (Appendix C: Table C1a). For juveniles, we found lower local survival for males than females (Fig. 2d; Appendix

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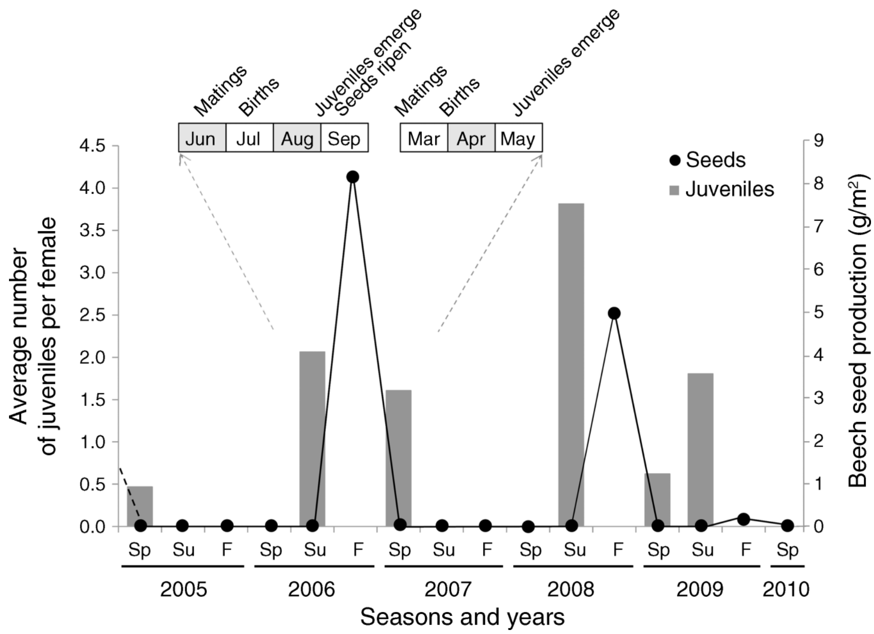

FIG. 1. Average number of juveniles produced by adult female eastern chipmunks (*Tamias striatus*) and estimated beech (*Fagus grandifolia*) seed production per season. Sp, Su, and F indicate spring, summer, and fall (i.e., autumn) calendar seasons, respectively. The black dashed line indicates that there was a mast in fall 2004, but we did not quantify beech seed production. Also presented is the timing of chipmunk reproduction preceding and following a mast.

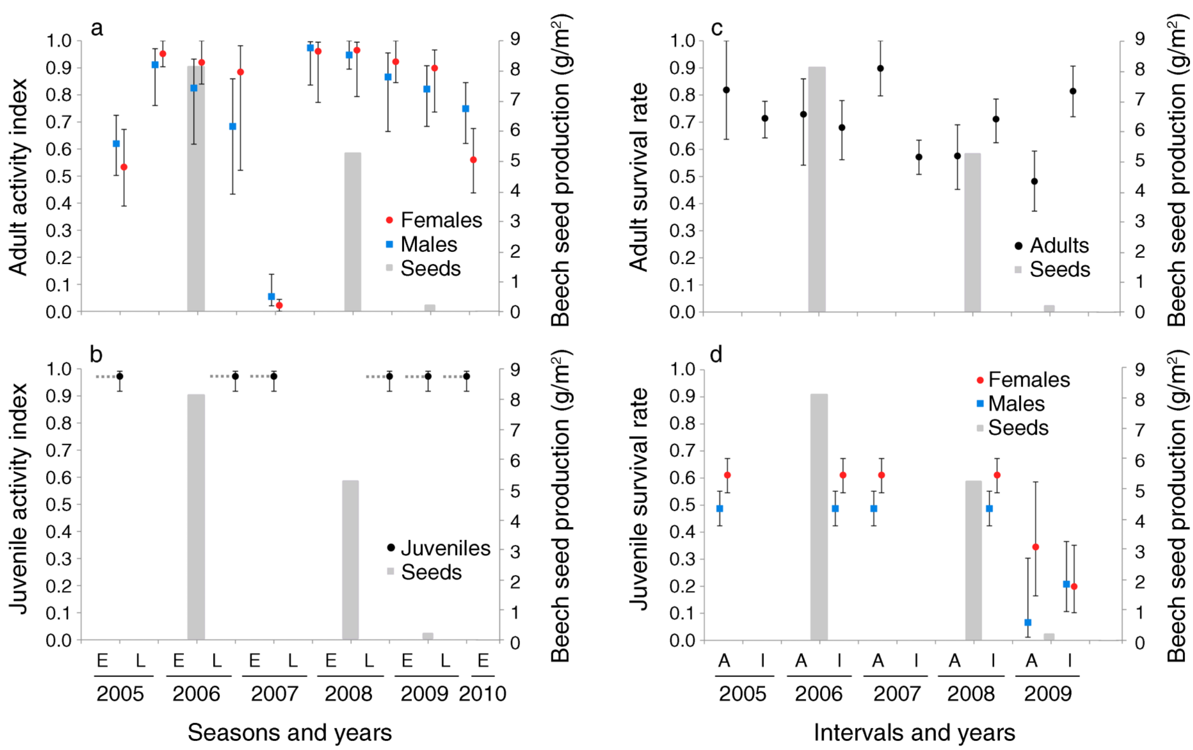

FIG. 2. Biannual (six-month) activity index based on recapture rates for (a) adults and (b) juveniles, and survival rates for (c) adults and (d) juveniles (mean with 95% CI) from the best CMR (capture–mark–recapture) models in relation to beech seed production (gray bars); see Appendix C. Active seasons are early (E, late April to 31 July) and late (L, 1 August to late October); intervals A and I indicate active (between E and L) and inactive (between L and E) periods during which survival rates are estimated. The dotted lines in panel (b) indicate that a single recapture rate is calculated at $t+1$ for each cohort born at time t.

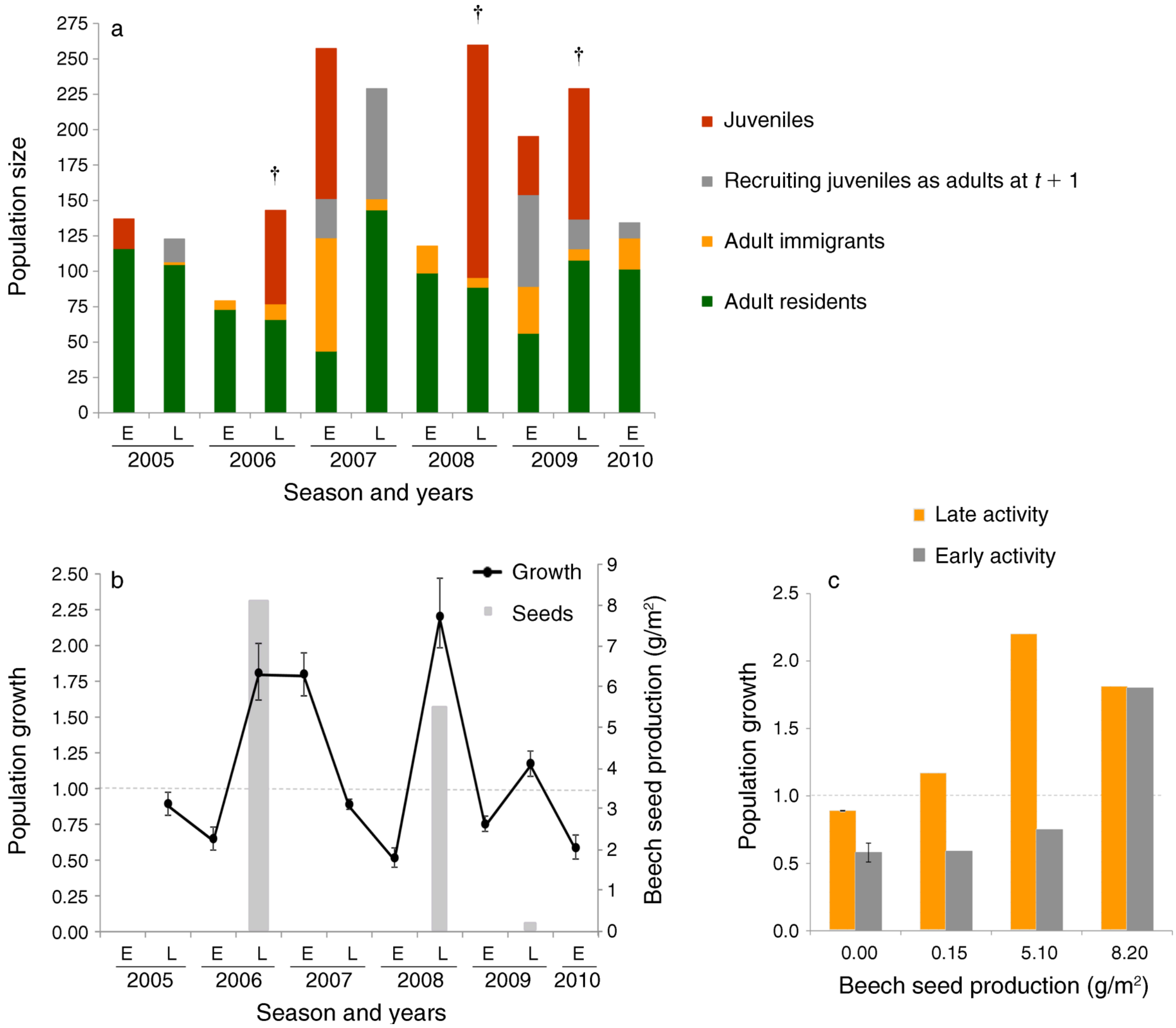

FIG. 3. (a) Relative contribution of juveniles, recruiting juveniles as adults at time $t+1$, adult immigrants, and adult residents to population size; dagger symbols (†) indicate seasons of beech seed production (Appendix B). Because recapture rates are <1.0, recruiting juvenile and resident adult numbers are estimated from the best CMR models (Appendix C). (b) Population growth (mean with 95% CI), measured as the ratio of population size at time $t+1$ divided by that at time t. E and L indicate early (late April to 31 July) and late (1 August to end of October) active seasons. Gray bars indicate beech seed production. (c) Relationship between mean population growth (±SE when $n > 1$) and beech seed production. Population size is stable when population growth = 1.0 (dashed line).

C: Table C1b). We also found that juvenile survival during winter 2009 was much lower than survival of the other cohorts, except for the 2009 active season (Fig. 2d; Appendix C: Tables C1b and c).

Population growth.—There was a threefold change in population size on the grid over the course of this study (highest: $n = 260$ in late active 2008; lowest: $n = 79$ in early active 2006; Fig. 3a). Accordingly, there were large fluctuations in population growth across seasons, ranging from 0.52 to 2.20 (Fig. 3b). There was a significant positive relationship between the number of juveniles and population growth (juveniles: 0.009 ± 0.002, $t = 5.722$, df = 1, $P > 0.001$, $r^2 = 0.78$; Fig. 3a; Appendix C: Fig. C2), but there was no relationship between population growth and the number of resident adults, immigrants, and recruiting juveniles to the site (all P values ≥ 0.76). Consequently, population growth increased significantly with beech seed production (0.11 ± 0.03, $t = 4.12$, df = 1, $P = 0.004$, $r^2 = 0.67$; Fig. 3c). Also, population growth was greater in the late active season than the early active season (0.53 ± 0.19, $t = 2.77$, df = 1, $P = 0.03$; Fig. 3c) but there was no interaction between season and beech seed production (season × seed: −0.007 ± 0.06, $t = -0.11$, df = 1, $P = 0.92$).

DISCUSSION

As expected, chipmunk population dynamics were tightly linked with beech seed production. Our results also support the hypothesis that chipmunks reproduced in anticipation of a mast, synchronizing juvenile emergence with the peak of seed availability. Our results suggest that the cues used by chipmunks to anticipate a mast could also influence the regulation of adult aboveground activity, but not juvenile activity. Between-year variation in adult survival was independent of beech seed production, whereas juvenile survival tended to be positively affected by beech seed availability. Finally, we found that chipmunks also closely tracked beech seed production by reproducing every spring following a beech mast. Although we acknowledge that site replication would have been useful to assess the extent of spatial variation in age-specific life-history responses, our goal here was to detail these adaptations using extensive, individual-based monitoring of hundreds of chipmunks over several years, which was logistically feasible for only one site.

Chipmunks anticipated beech masting by mating in June, synchronizing juvenile emergence with ripening of beech seeds in September. These patterns are similar to those observed in red squirrels (Boutin et al. 2006). Boutin et al. (2006) suggested that this behavior represents an adaptation of squirrels to counteract the predator satiation cycle imposed by masting plants. The predator satiation hypothesis suggests that lagged numerical response of consumer communities to pulse resources (Jones et al. 1998, Ostfeld and Keesing 2000, Yang et al. 2010) allows the resource to appear at low consumer densities, preventing them from consuming all of resource (Janzen 1971, Kelly and Sork 2002). Reproductive anticipation of a mast caused squirrels to reach peak population density just before spruce masting (Boutin et al. 2006). We also found that reproducing in anticipation of a mast substantially increased total chipmunk population size in September, when beech seeds are ripe and begin to fall to the forest floor. Similarly, anticipatory reproduction allowed juvenile white-footed mice to capitalize on high cicada abundance (Marcello et al. 2008), which represents an even more ephemeral resource than storable seeds.

We found that mast anticipation by chipmunks also extended to the regulation of their aboveground activity. During years of no mast, adults skipped summer reproduction and remained in their burrows from July until the next spring (see also Munro et al. 2008), despite the opportunity to forage on alternative food items until October, which juveniles capitalized on. Such behavior is surprising for animals having a rather short adult longevity of about 1.4 years (average adult annual survival rate = 0.48, adult longevity average estimated as $-1/\ln(\phi_{avg})$). Many consumer populations, responding to resource pulses, crash to extremely low numbers during inter-pulse intervals (White 2008, Yang et al. 2010). Yet, Ruf et al. (2006) reported that dormice (*Glis glis*) had higher survival rates when they reduced reproductive activity during beech failure years, compared to survival during mast years when they reproduced. Many species, subject to environmental cycles, inhibit reproduction and activity (e.g., diapause, dormancy, torpor) to favor survival until conditions are more favorable for juvenile production. Here we found that adult survival was independent of beech masting, suggesting that extreme suppression of aboveground activity and reproduction skipping may indeed buffer the expected survival costs associated with conditions of low resource availability. Anticipatory cues thus allowed chipmunks to regulate reproductive behavior and aboveground activity, causing their population to reach high density during mast years, thus potentially dampening predator satiation cycles.

It remains surprising that increasing reproductive effort prior to the availability of additional resources is energetically possible for small mammals. This response suggests that females normally pursue energetically conservative reproductive strategies, such that the prospects for juvenile survival are a more important determinant of variation in reproductive output than proximate energetic constraints affecting the capacity of females to successfully reproduce (Boutin et al. 2006). Alternatively, given the potential for nutrient (e.g., nitrogen), rather than energetic limits to reproduction and the high nutrient content of immature plant parts (e.g., seeds, buds) and immature insects, White (2007) and Dunn et al. (2011) suggest that increases in reproductive effort prior to resource pulses may simply reflect responses to high nutrient availability rather than anticipation of high energy availability. Yet, several lines of evidence make us believe that chipmunks express adaptive reproductive plasticity by using anticipatory reproductive effort during mast years and restrained reproduction during years of mast failure. First, chipmunks collect the vast majority of their food items on the forest floor (Elliott 1978) and thus have very limited access to immature seeds present in the forest canopy. However, they could have had access to some chemical cues expressed by flowers appearing in late April–May (Berger et al. 1981). Second, chipmunks store enormous amounts of seeds in their burrows (Humphries et al. 2002, Kuhn and Wall 2008), which serves to decouple resource availability in their larder hoard from resource availability on the forest floor. For example, chipmunks had enough food in their burrows in a non-mast year to survive underground from July to May and presumably could have used these reserves to support current reproduction instead of survival for future reproductive opportunities. Third, juveniles that are born in summer and emerge in August–September should be very sensitive to beech availability because they only have one or two months to find their own burrow and accumulate a larder hoard for winter. In many species, the resources accumulated by juveniles, either as fat or food storage, greatly affect winter survival (e.g., Gaillard et al. 2000, Larivée et al. 2010). During the summer 2009, chipmunks reproduced in anticipation of a mast, but beech seed production in autumn was very small, perhaps because chipmunks misread the cue to masting or that the masting event itself was impeded (e.g., by poor pollination). Interestingly, we observed a lower juvenile survival rate in the following winter, whereas adult survival remained unaffected. These results are consistent with resource anticipation being driven by juvenile survival prospects rather than reproduction being constrained by food availability for adults, but the latter remains a parsimonious alternative explanation made more attractive by the lack of a demonstrated proximate mechanism linking current reproduction to future resource availability.

Consistent with the common resource-tracking strategy (Yang et al. 2010), chipmunks also reproduced during early springs that followed beech masts. Juveniles emerged in spring and remained active during the entire season (until October). We previously found that juveniles that emerged in spring 2009 were able to mate in June of the same year (Bergeron et al. 2011). This timing of reproduction is commonly observed as a rapid resource-tracking strategy by rodents. For example, bank voles (*Myodes glareolus*) and mice (*Apodemus flavicollis*, *P. leucopus*, *P. maniculatus*) intensively reproduce during winters following a mast (Jensen 1982), causing these populations to reach outbreak levels in spring. In the absence of a mast, these rodents did not mate during winter. It is difficult to determine why chipmunks are not reproducing during early springs following mast failures, because juveniles that emerge would have an entire active season (six months) to locate a burrow and establish a larder hoard, typically under mast conditions, given the alternative-year mast cycle in our system. One possibility is that the larder hoards remaining in adult burrows in late winter and early spring may not be sufficient to support the pre-emergence euthermy required for reproductive preparation (Humphries et al. 2003) and subsequent energy demands invoked by early spring reproduction. Alternatively, adults may refrain from reproducing if emergent juveniles do not have access to beech seeds in the spring, left over on the forest floor from the previous autumn. For instance, Jensen (1982) found that *M. glareolus* and *A. flavicollis* only consumed 1–10% of the beech nuts produced during heavy mast years, meaning that many remaining seeds were available in the following spring.

Chipmunk reproduction in anticipation of and in response to tree masting appears to be effective in maximizing the use of pulsed resources by synchronizing peak reproductive effort with peak seed availability. The capacity of chipmunks to undergo a rapid demographic response during a mast, independently of adult density, suggests that extrinsic factors such as food abundance are key drivers of their population dynamics, which may reduce the strength of negative intrinsic factors such as density dependence. In this system, adults expressed high survival during mast failures, whereas juveniles benefitted greatly from mast production. Although fertility and juvenile recruitment are key factors driving the population dynamics of short-lived species (Heppell et al. 2000), our results suggest that pulsed resources can lead to conservative life histories favoring adult survival. Conservative strategies are usually observed in long-lived species in which adult survival is very high and relatively invariant, whereas juvenile survival is very sensitive to environmental variation (Gaillard et al. 2000, but see McAdam et al. 2007). Nevertheless, a combination of slow and fast life-history strategies may allow many short-lived species to make the most of a fluctuating environment.

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APPENDIX A

CMR (capture–mark–recapture) analyses: goodness-of-fit tests and modeling approach for the eastern chipmunk, *Tamias striatus* (*Ecological Archives* E092-177-A1).

APPENDIX B

Estimating population size (*Ecological Archives* E092-177-A2).

APPENDIX C

Table of the best-fitted models of survival rates and recaptures probabilities; figures on the relationship between observed and known-alive resident adults, and on the relationship between population growth and the number of juveniles (*Ecological Archives* E092-177-A3).

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